THESIS PROJECT
TITLE PROPOSAL
(CS0025 – Software Engineering 1)

Group Name: **Statisfied**	Program: **Bachelor of Science in Computer Science with specialization in Data Science**	
Member's Name: 1. Dawn Andrei Pamesa (Project Manager) 2. Rogelio Porras (Member 1) 3. John Ashley Alba (Member 2) 4. Ralph Blanco (Member 3)	Term: [✓] 1st [ ] 2nd [ ] 3rd	Academic Year: 2026 - 2027
Proposed Project Title 4: **Experimental Early Warning of Highland Vegetable Price Collapses at the Benguet Agri-Pinoy Trading Center**		

1.0 Area of Investigation:

Benguet is a major source of highland vegetables, and the Benguet Agri-Pinoy Trading Center (BAPTC) in La Trinidad is described by the Department of Agriculture as the country's biggest vegetable trading facility (Department of Agriculture, n.d.). It traded nearly 170,000 metric tons of produce in 2020, up from 269 metric tons when it started trading (Ocampo, 2021). Vegetable prices are sensitive to supply (Hinlo & Lee, 2026; Qiao et al., 2023). When many farms harvest the same vegetable in the same weeks, the price can fall sharply, and part of the harvest can go unsold. Carrots are an example. Antolin et al. (2024) list low buying prices in the peak season among the problems of the Benguet carrot industry. In one tracked lot of Benguet carrots, they measured total losses of about 35% after five to six days and about 45% after seven to eight days at the wholesaler and retailer level. A recent example is close at hand: the Department of Agriculture Cordillera Administrative Region (DA-CAR) and BAPTC sold at least 1,270 kilograms of previously unsold carrots through a joint market intervention (Department of Agriculture–Cordillera Administrative Region [DA-CAR], 2026).

This study asks one question: can the weekly price of a highland vegetable at BAPTC, and the risk

of a price collapse, be forecast early enough for BAPTC, DA-CAR and farmers to act? The study separates two kinds of events. A price-collapse event is a week in which the price falls far below its usual level for that time of the year. A supply-glut event is a price-collapse week that also has unusually high trading volume. The price records are daily, and whether the DA-CAR file also has volume is not yet known, so the main target is the price-collapse event, and supply-glut events are evaluated only if volume is received. The study starts with carrots, because they have the most recent glut evidence, and extends to the other highland vegetables that the data covers.

BAPTC is the intended client. It was set up by the Department of Agriculture, which announced plans in 2021 to spin it into a corporation (Department of Agriculture, n.d.; Ocampo, 2021). DA-CAR, the Department's Cordillera office, told the group by phone that it can provide about five years of daily vegetable prices in Excel, with more than 5,000 rows, and can pass the group's request on to BAPTC. BAPTC has also allowed a site visit. The group will confirm which vegetables and which years the file covers, and whether it also has volume or grade columns, once it is received. The Philippine Information Agency reports that the Department of Agriculture plans digital trading and profiling at BAPTC (Philippine Information Agency, 2025), so the study fits a direction that the agency has already announced.

The group's reading of the center's three main problems, and the one chosen:

Problem	Evidence	Status
1. Prices collapse during gluts and produce goes unsold	Unsold carrots had to be sold through a joint DA-CAR and BAPTC intervention (DA-CAR, 2026).	CHOSEN as the main problem. Severity ranking to be confirmed in the client interview.
2. Price and volume information is scattered	The group found BAPTC prices posted mainly on a social media page, and the group could not reach the center's own web portal when it checked in September 2026.	Handled by one cleaned, time-ordered data set. Severity ranking to be confirmed in the client interview.
3. No forward-looking signal for harvest and selling decisions	The information the group found is about past prices. The Department of Agriculture plans digital trading and profiling at BAPTC (Philippine Information Agency, 2025).	Handled by the forecast and the alert. Severity ranking to be confirmed in the client interview.

The analytics type is predictive (forecasting price and price-collapse risk), with descriptive summaries of past price collapses. The client's preferred type will be confirmed in the client interview.

The proposed system has four parts:

- Data preparation: cleans and combines the price and volume records into one weekly data set per vegetable.
- Forecasting: predicts the price one to four weeks ahead; model F also gives a low, a middle and a high value.
- Price-collapse alert: warns when the model's alert score, the low value in the case of model F, falls below the price-collapse threshold.
- Dashboard: shows the forecasts, the ranges where a model provides them, the alerts and how accurate past forecasts were, for BAPTC and DA-CAR staff.

Algorithm/Technique Justification

Identified problem	Price collapse and unsold produce during supply gluts, with no early signal for BAPTC, DA-CAR or farmers.
Computational task	Time-series forecasting (regression) of the weekly price one to four weeks ahead, and classification of price-collapse weeks, which are rare events.
Proposed technique	To be confirmed by experiment. The leading candidate is XGBoost with a quantile (pinball) loss at the 10th, 50th and 90th percentiles. The low value gives the early warning, and a validation-tuned alert margin (chosen by F2, which favors recall) turns it into an alert. No new algorithm is claimed, and the final technique is the one the experiments support.
Comparison	Last observed price, seasonal naive, SARIMA, Random Forest, and XGBoost with the standard squared-error loss, all described in 2.0.
Required data	About five years of daily price records per vegetable from DA-CAR (more than 5,000 rows, as told to the group), with volume where it exists. The adviser's rule of at least 5,000 usable rows from 2021 onward is checked when the file arrives.
Expected output	A price forecast for one to four weeks ahead (with a range from model F), a price-collapse alert, accuracy results for each vegetable, and a dashboard.
Evaluation	Rolling-origin testing over several later windows and an untouched final period. MAE, RMSE and MASE for the price; range coverage; precision, recall, F2, PR-AUC, lead time and alert share for the alerts; training time.

2.0 Background and Rationale of the Project/System:

Vegetable prices are hard to forecast because the produce is perishable, seasonal and sensitive to supply. Hinlo and Lee (2026) split Philippine onion and garlic price variance into demand, supply and import parts. They found that demand variability was the largest factor, followed by supply, and

that seasonal production made the swings bigger. Qiao, Kang and Ahn (2023) did the same for five Korean vegetables, including cabbage, and found that domestic production was the main driver of price changes.

Studies apply many methods to vegetable prices. Mao et al. (2022) used an ARIMA model on more than 51,000 vegetable prices from 2020. Paul et al. (2022) compared neural network, support vector, random forest and gradient boosting models for brinjal wholesale prices in seventeen markets in Odisha, India. Other studies use deep learning with extra variables: Nayak et al. (2024) added weather to price models for tomato, onion and potato; Bao et al. (2025) added a search index for extreme weather and an economic policy uncertainty index to a CNN-LSTM-Attention model for Chinese vegetable prices; Zhang et al. (2024) trained an LSTM on daily prices of six vegetables in Beijing; Wang et al. (2025) combined decomposition, an optimizer and a GRU for six vegetables; and Choi et al. (2023) forecast Korean garlic prices with an attention-based hybrid.

Tree-based methods are often strong when the data is a table of lagged values and calendar features. Haselbeck et al. (2022) compared nine machine learning methods and three classical methods for horticultural sales and found that XGBoost was the best in 14 of 15 comparisons. Li et al. (2024) joined Random Forest and XGBoost to forecast short-cycle agricultural sales and reduced the MAPE by 10% and 12% compared with each model alone. Zhao et al. (2025) joined a temporal convolutional network and XGBoost to forecast rice, wheat and corn prices.

The studies closest to this one are about carrots and the Philippine highlands. Mao et al. (2022) applied an ARIMA model to the 2020 prices of three vegetables, cabbage, carrot and eggplant. Bulut (2025) used daily prices and trade volumes from the Turkish Ministry of Commerce for parsley, lemon, cauliflower and carrot, and compared CNN, LSTM and Random Forest models on 10-day price forecasts. Lubag et al. (2025) built a model for hinterland lettuce selling prices with recurrent neural networks.

These studies forecast a price or a sales figure. In the studies the group found, none turns the forecast into a price-collapse or glut alert, none gives a price range that can be used for a warning, and none uses BAPTC trading records for carrots. Mao et al. (2022) and Bulut (2025) show that carrot prices can be forecast, the first with ARIMA and the second with trade volume as an input, but with Chinese and Turkish data and without an alert. Neither used XGBoost. Models C and D below use the same families that were used on carrots in those studies, ARIMA (as SARIMA) and Random Forest, so the XGBoost models E and F can be compared with them directly. The group searched Crossref, OpenAlex and its own literature notes; the search was not exhaustive.

Gradient boosting can predict a range instead of one number by using a quantile (pinball) loss. Bauer et al. (2024) propose boosting for regression quantiles with the pinball loss, and Laimighofer et al. (2022) put an expectile loss into XGBoost to give extra weight to the lowest values when predicting low river flows. Masnadi-Shirazi and Vasconcelos (2011) describe cost-sensitive boosting, where one kind of error is made more costly than another. At BAPTC the costly error is missing a price collapse. This study does not claim to introduce a new gradient boosting algorithm.

It applies XGBoost (Chen & Guestrin, 2016) with a quantile loss, which changes its objective, to estimate a low, a middle and a high price forecast. A validation-tuned alert margin then turns the low forecast into an early-warning signal. The margin is a decision rule applied after training and is not itself a modification of XGBoost. The contribution is the evaluation of this forecasting and alerting approach on BAPTC vegetable prices, starting with carrots, against simpler options.

Task and data. For each vegetable the data is a weekly series of the wholesale price, and of the volume if the file has it. A daily file is averaged to weeks. The forecast horizon is one to four weeks. Each model sees only information that was known at the forecast date.

Event definitions. A price-collapse event is a week in which the price is below the 25th percentile for that time of the year, calculated from the training years only. A supply-glut event is a price-collapse week in which the volume is also above the 75th percentile for that time of the year. The two are evaluated separately. "That time of the year" means a seasonal group: the target week and the two weeks on each side of it, pooled over the training years only. The grouping and the percentile method are fixed before the final evaluation, so that single calendar weeks do not give unstable estimates and nothing from the final period leaks in. If the file has no volume, only price-collapse events are evaluated and no supply-glut claim is made. The percentiles are a starting rule. BAPTC and DA-CAR staff will be asked whether they match what they call a glut, and the rule changes if they say so.

Models compared. A is the last observed price. B is the price of the same week last year (seasonal naive). C is SARIMA (Hyndman & Athanasopoulos, 2021). D is Random Forest (Breiman, 2001). E is XGBoost with the standard squared-error loss (Chen & Guestrin, 2016). F is the leading candidate: XGBoost with the pinball loss at the 10th, 50th and 90th percentiles. It leads because of the studies above, not because it is confirmed. All six models are candidates, and the final technique and its alert settings are selected using validation results only. The untouched final period evaluates the selected technique and reports whether its performance supports the selection, including negative or inconclusive findings. All candidate models produce price forecasts and alert scores. Model F also produces lower, median and upper quantile forecasts, which are evaluated separately, and ranges are shown only for models that produce valid uncertainty estimates. Models D to F use the same features: lagged prices and volumes, rolling averages, week of the year, month and holidays. Weather data is an optional extra if time allows.

How the alert works. Each model gives an alert score: the predicted price for A to E, and the predicted 10th percentile price for F. An alert is raised when the score is below the price-collapse threshold plus a margin. The study compares complete forecasting-and-alerting approaches and does not treat point forecasts and lower-quantile forecasts as the same quantity. Each method's alert margin is chosen separately on validation data only, with the same protocol (the margin that gives the highest F2). The final test reports the share of weeks that receive an alert next to precision, recall, F2 and PR-AUC, so that a method is not credited for simply alerting more often. F2 is a proposed priority: missing a price collapse is assumed to cost farmers more than a false alarm, and

the actual costs will be confirmed with BAPTC staff.

Training and testing. Each vegetable is forecast at horizons of one, two, three and four weeks, with a separate model for each horizon. At each forecast origin, which is the last week with known data, a model is trained only on observations available by that date and then predicts the price at the target week. For each horizon, a training example is used only if the price of its target week has already been observed by the training cutoff. The origin moves forward week by week (rolling origin) with an expanding training window. Lagged features, seasonal percentiles and any scaling are calculated from the training weeks only. Hyperparameters and the alert margin are fixed on validation weeks before the last year of data, which is kept untouched until the final evaluation. Random splitting is not used. Results are reported for each vegetable and each horizon.

Metrics. Price: mean absolute error (MAE), root mean squared error (RMSE) and MASE, which compares the error with the seasonal naive forecast. Range (model F only): the share of actual prices inside the 10th to 90th percentile range (80% is the calibration target, not a guaranteed result), the average width of that range, the pinball loss, and whether the predicted quantiles stay in the right order. Alerts: precision, recall, F2, the area under the precision-recall curve (PR-AUC) and lead time, meaning the number of weeks between the first alert and the price-collapse week, plus the share of weeks that receive an alert. Efficiency: training and prediction time. The metrics are to be agreed with the adviser and BAPTC staff.

Statistical check. The difference between F and E, and between F and B, in MAE and in F2 is reported with a bootstrap confidence interval that resamples blocks of four weeks, so that a small gain is not presented as a win. Four weeks matches the longest forecast horizon, because forecasts made from nearby origins have overlapping errors, and blocks of two and eight weeks are tried as a check. With few collapse weeks the intervals will be wide, and the results say so.

Rare events. Price-collapse events are rare, so the number of such weeks for each vegetable is reported first. If a vegetable has very few, its alert results are reported as descriptive only.

The dashboard shows the forecast, its range, the alert and how accurate past forecasts were. It is a decision-support tool. BAPTC and DA-CAR staff make the final decision.

3.0 Statement of Objectives:

General objective. To develop and evaluate an early-warning approach for highland vegetable price collapses at BAPTC by comparing six candidate forecasting models on records from DA-CAR and BAPTC, with XGBoost with a quantile loss and a recall-focused alert rule as the leading candidate, and with supply-glut analysis done separately when trading-volume data are available.

Specific objectives. The specific objectives are as follows:

1. Obtain about five years of price records, and volume records if they exist, from DA-CAR or

BAPTC, check that they hold at least 5,000 usable rows from 2021 onward across the vegetables, note how many rows each vegetable has, and prepare a clean weekly data set with documented cleaning rules before any modelling starts.

2. Define price-collapse events, and supply-glut events if volume is received, with BAPTC and DA-CAR staff using the percentile rules in 2.0, and report the number of such weeks for each vegetable.
3. Build the baseline models A to E and measure their price error and alert results on rolling-origin windows.
4. Build model F and choose its alert margin by the highest F2 on validation data, before the final test.
5. Evaluate whether model F achieves a higher F2 than the strongest baseline among A to E, selected on validation data and then fixed, on the untouched final period for carrots, and report the same comparison for the other vegetables, including negative or inconclusive results. The final technique and its alert settings are selected on validation results only, and the final period is used only to evaluate them.
6. Evaluate the 10th to 90th percentile ranges of model F by how often they contain the actual price (80% is the target), their average width and whether the quantiles stay in order, and report this for each vegetable.
7. Build a dashboard of forecasts, ranges and alerts, and test it with BAPTC or DA-CAR staff if they agree to take part, or with representative users if not, and say which.

4.0 Scope and Limitations of the Study:

- One market: BAPTC in La Trinidad. Carrots come first; the other vegetables are included only if the data covers them.
- Weekly forecasts one to four weeks ahead, using aggregated price and volume records. No individual farmer or trader information is used or needed.
- Data comes from DA-CAR, and from BAPTC through DA-CAR or a site visit. The group does not scrape social media pages for data.
- The model gives a forecast and an alert. It does not decide what BAPTC or farmers should do, and it is not deployed live. The group hands over a working prototype.
- The data has not been received yet. DA-CAR says it holds daily prices with more than 5,000 rows in total. About five years of daily data give about 1,800 rows for each vegetable, so the rows of different vegetables are not treated as one long series. If the file has no volume records, only price-collapse events are evaluated. If fewer than 5,000 usable rows remain from 2021 onward after cleaning, or the history of a vegetable is too short for reliable forecasting, the study narrows its scope to the vegetables with enough history, and says so. Rows from different

vegetables are not treated as one continuous time series.

- Five years of data may hold only a few price-collapse weeks for each vegetable, so alert results can be uncertain. The study reports the number of price-collapse weeks, uses blocks of weeks in its bootstrap, and does not claim more than the data supports.
- Planting plans, weather and transport disruptions are not part of the main model. Weather is an optional extra. The price-collapse and glut percentiles are a starting rule that BAPTC and DA-CAR staff may change.
- Forecasts are decision support and not guarantees. The dashboard shows ranges and past accuracy so that nobody reads a single number as certain.

5.0 Target Beneficiaries:

BAPTC management and staff, through an early warning of price collapse and a clean, structured price history.

DA-CAR staff, who help BAPTC and farmers sell unsold produce, through earlier notice of likely price collapses.

Benguet vegetable farmers and traders, who can plan harvest and selling with a forward signal, reached through DA-CAR and BAPTC.

Researchers and future students, who get a documented method and results for a highland vegetable market in the Philippines.

6.0 SDG Category Number:



SDG Number: 2

SDG Title: End hunger, achieve food security and improved nutrition and promote sustainable agriculture

Justification: Target 2.3 aims to double the agricultural productivity and incomes of small-scale food producers, including through secure and equal access to markets. An early warning of price collapse could help Benguet farmers and BAPTC decide when to harvest, sell or move produce before prices fall sharply. The project also supports target 12.3, which aims to reduce food losses along production and supply chains, including post-harvest losses, because produce that is moved earlier is less likely to go unsold. The study does not measure that impact. It evaluates the quality of the forecasts and alerts, and any benefit to farmers would need a later study.

7.0 References:

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Approved by:

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Panel Member 1 / Date

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Panel Member 2 / Date

Noted by:

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Thesis 1 Course Adviser / Date

1. AI Interaction Log (if applicable)

If your group used any Generative AI tool while completing the assessment, document how the tool was used. The log should demonstrate that AI supported the group's work rather than replacing the group's research, analysis, and decision-making.

Group Member	AI Tool Used	Prompt / Purpose	AI Output Used For	How the Output Was Verified or Modified
Dawn Andrei Pamesa	Claude (Claude Code)	Read the group's earlier BAPTC plan and literature notes; checked every cited paper on Crossref and read its abstract; drafted the sections and the method design; formatted the document in the official layout.	Drafts of each section, reference checks, and formatting.	Every DOI was checked on Crossref and each statement about a study was matched to its abstract. The text was rewritten in the members' own words and checked against the sources.
	Custom GPT (thesis manuscript checker)	The draft was pasted in for adviser-style reviews in two rounds and revised from its feedback.	Feedback on the method design, the evaluation, the wording and missing items.	The group checked each suggestion, did its own research on it, and cross-referenced it before applying it, and chose the best option.
	ChatGPT	Helped find sources.	A list of possible sources.	The sources were cross-checked.
Rogelio Porras	ChatGPT and Codex	Helped search for sources, and helped brainstorm and proofread his writing.	A list of possible sources, brainstorming ideas, and proofreading.	He read the sources himself before using them.

John Ashley Alba	ChatGPT	Helped gather sources and ideas, and helped check his work.	Suggestions of sources and ideas, and checking his work.	He read the sources himself and cross-checked them with other sites.
Ralph Blanco	None	None	None	None

2. AI Usage Declaration

We declare that any use of Generative AI in completing this Technical Assessment has been properly disclosed in the AI Interaction Log. We understand that AI is used only as an assistive tool and that our group remains responsible for verifying the accuracy, relevance, credibility, and originality of the submitted work.

We confirm that the final title proposal, objectives, scope, algorithm justification, and research outputs reflect our group's understanding and decisions. All literature, studies, and other external sources used have been properly cited and verified.

Group Name: Statisfied

Date: October 31, 2026

Group Member	**Signature**
Dawn Andrei Pamesa	
Rogelio Porras	
John Ashley Alba	
Ralph Blanco	

3. Team Contribution Matrix

The total contribution percentage must equal 100%.

Group Member	Assigned Task/Responsibility	Specific Contribution/Output	Contribution (%)
Dawn Andrei Pamesa	Project Manager; contacted BAPTC and DA-CAR as the client; planning with Ralph.	BAPTC and DA-CAR client contact; planning and method design with Ralph; the proposal text, done with Claude and the custom GPT.	30%
Rogelio Porras	Review of related literature and systems (RRL and RRS).	Sources and write-up for the RRL and RRS.	23%
John Ashley Alba	Review of related literature and systems (RRL and RRS).	Sources and write-up for the RRL and RRS.	23%
Ralph Blanco	Planning and brainstorming with the Project Manager.	Ideas and planning input for the proposal.	24%
		Total	100%